

u 2s



Chapter 4-5 Application/Communication

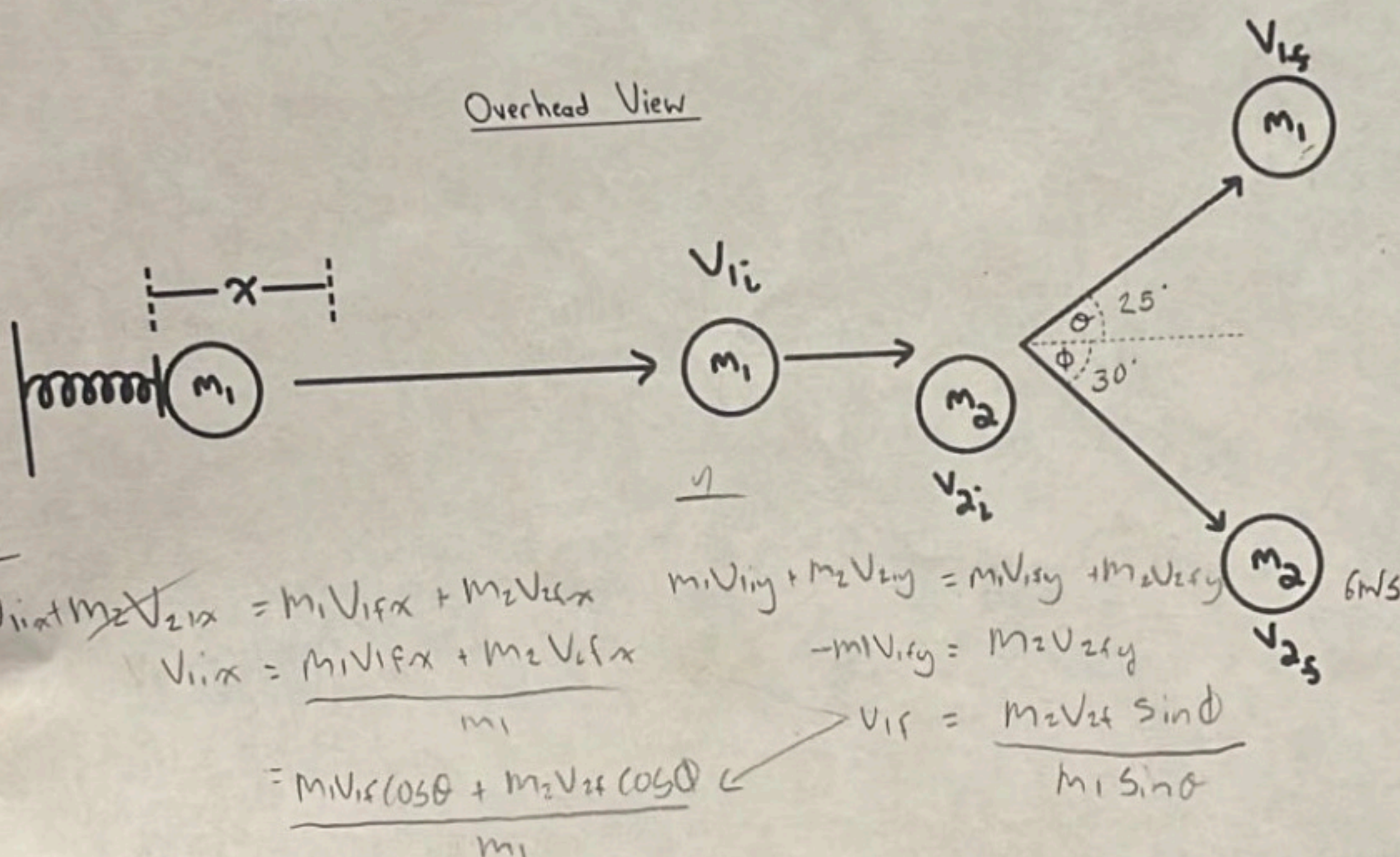
Application (8 marks)

1. A mass m_1 is held in front of a compressed spring. When m_1 is released it travels forward until it hits a stationary mass, m_2 . The collision is a glancing collision as shown in the diagram. Using the following values, calculate the compression of the spring. There is no friction. (8 marks)

$$m_1 = 5.00 \text{ kg} \quad m_2 = 2.00 \text{ kg} \quad v_{2i} = 6.00 \text{ m/s} \quad k = 52.8 \text{ N/m} \quad \theta = 25.0^\circ \quad \phi = 30.0^\circ$$

No numbers until the very end! Components are a good place to start...

Overhead View



$$m_1 v_{1i} + m_2 v_{2i} = m_1 v_{1f} + m_2 v_{2f}$$

$$v_{1i} = \frac{m_1 v_{1f} + m_2 v_{2f}}{m_1}$$

$$= \frac{m_1 v_{1f} \cos \theta + m_2 v_{2f} \cos \phi}{m_1}$$

$$v_{1f} = \frac{m_2 v_{2f} \sin \phi}{m_1 \sin \theta}$$

$$v_{1i} = \frac{m_1 \cos \theta \left(\frac{m_2 v_{2f} \sin \phi}{m_1 \sin \theta} \right) + m_2 v_{2f} \cos \phi}{m_1}$$

$$E_{mi} = E_{mf}$$

$$\frac{1}{2} k x^2 = \frac{1}{2} m_1 v^2$$

$$x = \sqrt{\frac{m_1 v^2}{k}}$$

$$x = \sqrt{\frac{m_1 \left(\frac{\cos \theta \left(\frac{m_2 v_{2f} \sin \phi}{\sin \theta} \right) + m_2 v_{2f} \cos \phi}{m_1} \right)^2}{k}}$$

$$= \sqrt{\frac{5 \text{ kg} \left(\frac{\cos 25 \left(\frac{2 \text{ kg} \cdot 6 \text{ m/s} \cdot \sin 30}{\sin 25} \right) + (2 \text{ kg})(6 \text{ m/s})(\cos 30)}{5 \text{ kg}} \right)^2}{52.8 \text{ N/m}}}$$

$$x = 1.43 \text{ m}$$



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Communication (9 marks)/Knowledge (3 marks)

1. Pick one of the following and use equations and full sentences to explain your response. Include hypothetical numbers in your example. (C: /3 K: /1)
- Choose a hypothetical piece of sporting equipment and explain how impulse is used to improve performance.
 - Create a hypothetical case study to explain how impulse is used to increase safety.

When testing the safety of highway barriers concrete barriers and water filled ones were tested. The same model of car hits the barriers with an impulse of 19000 N·s. When the car hits concrete it is in contact for 0.5s and when it hits the barrier filled with water time of contact is 0.9s.

$$F_{\text{concrete}} = \frac{\Delta p}{t}$$

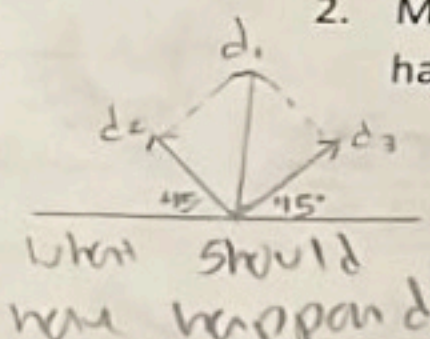
$$F = 38000 \text{ N}$$

$$F_{\text{water}} = \frac{\Delta p}{t}$$

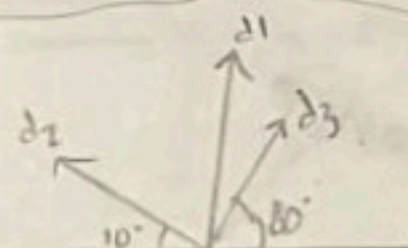
$$= 21111.1 \text{ N}$$

As seen from these values impulse helps us design safety objects as now we know that greater time of contact results in less magnitude of force. This is one of the reasons barrels filled with water are used.

2. Making reference to the 2D collision lab, use momentum or displacement vectors to show what SHOULD have happened and what actually did happen. What could have caused the variance? (C: /3 K: /1)



In 2-D collisions what should have happened is d_1 travels the furthest distance in a straight line perpendicular to the initial direction. d_2 & d_3 are the displacements of the glancing collision and if added should = d_1 . If momentum is conserved the velocity and initial displacement should be equal.



In our lab when we added d_2 & d_3 the answer was close, but not exactly d_1 . The variance could be caused by the angle of contact not being perfectly 45°. If this happens the resultant is on an angle which is why it might be more than d_1 . If momentum was conserved the angles should be equal. Also, if the ball was not perfectly round the actual displacement might increase. Additionally, taking the average of the points may not have been accurate. Considering each trial individually may have led to more accurate results.

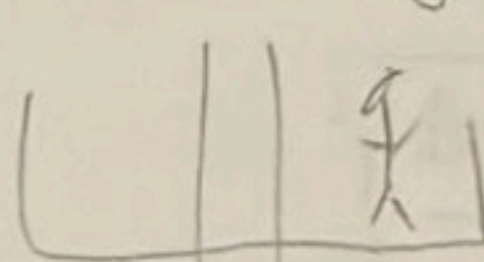
3. In this chapter we looked at non-conservative forces. Create your own example to help explain what work done by non-conservative forces means. Include a formula that can be used to calculate work done by non-conservative forces. (C: /3 K: /1)

In emergency situations, fire fighters need to use the fastest methods possible to leave and report to accidents. In many cases, sliding down the fire pole is a necessity.

Using conservative forces

Assume no friction. Ex
energy is fully conserved
 $E_{mi} = E_{mf}$
 $E_{gi} = E_{kf}$
 $mgh = \frac{1}{2}mv^2$

Using non conservative forces



$$K_i + W_{nc} = \Delta E_k \quad \Sigma W = \Delta E_k$$

$$W_g + W_{nc} = \Delta E_k$$

$$(mgh_i - mgh_f) + W_{nc} = \frac{1}{2}mv_i^2 - \frac{1}{2}mv_f^2$$

$$W_{nc} = (\frac{1}{2}mv_f^2 - \frac{1}{2}mv_i^2) - (mgh_i - mgh_f)$$

$$W_{nc} = (\frac{1}{2}mv_f^2 - \frac{1}{2}mv_i^2) + (mgh_f - mgh_i)$$



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